Promise Osaine Ekpo

Optimization, Reinforcement Learning, and Model Evaluation

Anticipated graduation December 2027. Available for a research internship, Summer 2027.

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Research Focus

I develop optimization based learning algorithms and evaluate their behavior through controlled, reproducible experiments. My work combines constrained multi agent reinforcement learning, inverse optimization, probabilistic modeling, and language model evaluation. I build training and evaluation systems in Python and PyTorch, connecting mathematical formulations to empirical tests. I am interested in extending this work to continual learning, adaptation, and training stability in deep networks.

Education

Cornell University, Ph.D. Computer Science. Advisor Prof. Angelique Taylor. GPA 3.94/4.00. *Aug 2023 to Dec 2027 expected*
Princeton University, M.S.E. Computer Science. Advisor Prof. Jaime Fernández Fisac. GPA 3.67/4.00. *Aug 2021 to May 2023*
University of Benin, B.Eng. Computer Engineering. GPA 4.77/5.00. Best Graduating Student, Dean's Honor Award. *Sep 2015 to Mar 2020*

Publications

Google Scholar

Peer reviewed and accepted

- **Ekpo, P.**, Agarwal, S., Grimm, F., Molu, L., Taylor, A. AdaFair MARL: Enforcing Adaptive Fairness Constraints in Multi Agent Reinforcement Learning. Poster accepted, *Women in Machine Learning Workshop at NeurIPS*, 2026.
- **Ekpo, P.**, Taylor, A., Molu, L. A Generalized Nash Equilibrium Seeking Scheme for Trauma Resuscitation. *IFAC World Congress*, 2026.
- Ogundare, O., **Ekpo, P.**, Wiggins, N. Ladder Logic Translation using Large Language Models in Industrial Automation. Accepted, *INFUS 2026* (Springer proceedings).
- Batra, Y., Pioldi, G., **Ekpo, P.**, et al. RFM HRI: A Multimodal Dataset of Medical Robot Failure, User Reaction, and Recovery Preferences. Accepted, *ACM Transactions on Human Robot Interaction*, 2026.
- Ogundare, O., Owadokun, T., Ogundare, T., **Ekpo, P.**, et al. Integrated Artificial Intelligence in Healthcare and the Patient's Experience of Care. *Scientific Reports*, 2025.
- Tanjim, T., **Ekpo, P.**, Cao, H., et al. Human Robot Teaming Field Deployments: A Comparison Between Verbal and Nonverbal Communication. *IEEE RO-MAN*, 2025, pp. 1699–1704.

Manuscripts and preprints

- **Ekpo, P.**, Jain, T., Sayatqyzy, A., Sil, S., Taylor, A. Toward Reducing Knowledge Induced Performance Gaps with Expertise Conditioned Inverse Semantics. Under review, *ACM/IEEE HRI 2027*.
- **Ekpo, P.**, Vijay, T., Mandalik, D., Jain, T., Ibrayeva, A., Sil, S., Tellex, S., Taylor, A. Toward Human-in-the-Loop Robot Failure Recovery: Bridging Communication Gaps in Human-Robot Collaboration. Under review, *ACM CHI 2027*.
- Pioldi, G., Batra, Y., Bai, Y., **Ekpo, P.**, Taylor, A. REPAIR Bench: A Benchmark for Robot Error Perception and Interaction Recovery. Preprint, 2026.

Selected Research

MARLHospital Simulator and Skill Aligned Fairness 2023 to Present
AIRLab, Cornell Tech

- Developed AdaFair MARL, a constrained cooperative learning framework. Derived a second order cone representation of the Jain fairness constraint and implemented primal-dual training with adaptive Lagrange multiplier updates.
- Built MARLHospital for heterogeneous agent coordination, fatigue, and shared tasks. Implemented QMIX, MAPPO, VDN, IQL, and IPPO in PyTorch and Stable Baselines3, scaling reproducible evaluation to more than 200 concurrent multi GPU Slurm jobs.
- Evaluated fairness and team performance against fixed penalty and reward shaping baselines. AdaFair MARL achieved 0.99 to 1.00 constraint satisfaction in the reported experiments while improving workload balance.

Vision Language Models (VLMs) as Inverse Optimization Designers 2026 to Present
AIRLab, Cornell Tech

- Implemented active set inverse optimization under a simplex constraint, reducing navigation weight recovery error from 0.190 to 0.004 with 20 demonstrations, versus 0.057 for maximum entropy inverse reinforcement learning.
- Designed controlled tests separating scene information from in context examples. Opposing examples changed recovered weights by 1.116, while changing the scene with examples fixed produced no change, diagnosing example copying by the VLM.
- Evaluated the pipeline over 12 scenes, six matched navigation pairs, 466 robot person encounters, and 33 checks.

Expertise Conditioned Inverse Semantics for Robot Help Requests 2026 to Present
AIRLab, Cornell Tech

- Developed a probabilistic listener model with latent knowledge, online belief updates, and request selection based on predicted human task success and communication cost.
- Froze the model on 272 outcomes from 49 listeners and evaluated without refitting on 80 later trials from 10 new listeners and an unseen task. Held out ROC AUC was 0.655 versus 0.458 for an utterance and task baseline.
- Quantified uncertainty with a paired listener bootstrap, obtaining an AUC improvement of 0.192 with a 95 percent interval of 0.028 to 0.320. Diagnosed limitations in task representations before fitting listener models.

Additional Research

Robot Failure Detection and Interaction Recovery 2024 to Present
AIRLab, Cornell Tech

- Built multimodal failure detection and temporal localization models using 214 interaction trials with 41 participants, achieving strict failure F1 of 0.80 versus 0.68 and median localization error of 2.97 seconds.
- Fine tuned Mistral 7B with QLoRA for recovery preference prediction, achieving Hit@5 of 0.76. Developed RFM HRI and contributed to REPAIR Bench, connecting model evaluation to human preferred recovery strategies.

Safe and Personalized Autonomy Research 2021 to 2023
Safe Robotics Lab and Beehive Research Laboratory, Princeton University

- Used a game theoretic framework to evaluate persuasiveness and safety in GPT 3 and GPT 4, measuring belief shifts of 12 to 50 percent toward false statements. Trained language models with PPO over 5,000 iterations.
- Used multi group Gaussian processes and fitted Q iteration to study personalized healthcare policies across patient populations.

Listener Differences in Human Robot Interaction, a Game, Dataset, and Benchmark 2025 to Present
AIRLab, Cornell Tech

- Built a web experiment platform, dataset, and language model benchmarks. Across 255 matched Expert and Novice request pairs, measured a 15.7 percentage point success gap, exact McNemar $p < .001$, using reproducible logs and controlled comparisons.

Industry Experience

Multimodal Safety Monitor and Explainable AI, Project Titanium Summer 2023
Siemens, Robust Intelligence Team

- Built a Python safety monitor fusing vision and real time location sensing. Applied GNNExplainer to graph attention networks for interpretable medical diagnosis, and presented both systems at the final review.

Data Engineer 2020 to 2021
Infinion Technologies, Microsoft Gold Partner

- Contributed to the Union Bank data warehouse migration, designed data standards, and built production ETL processes with SQL Server Integration Services and SQL.

Technical Skills

Learning and optimization. PyTorch, JAX, reinforcement learning, multi agent learning, constrained and inverse optimization, primal-dual methods, Gaussian processes, inverse semantics, QLoRA, LLM and VLM evaluation
Experiments and systems. Python, NumPy, pandas, scikit-learn, SQL, PostgreSQL, Linux, Git, Slurm, multi GPU experiments, automated validation, controlled comparisons, bootstrap confidence intervals, reproducible training and evaluation pipelines

Selected Talks

AdaFair MARL: Enforcing Adaptive Fairness Constraints in Multi Agent Reinforcement Learning. Northeast Controls Workshop 2026
A Generalized Nash Equilibrium Seeking Scheme for Trauma Resuscitation. IFAC World Congress poster 2026
MARLHospital, a multi agent reinforcement learning environment for medical decision making. Tapia Conference 2025

Honors and Awards

IEEE Robotics and Automation Society SPRISE Competition, Third Place, IROS 2026
Grace Hopper Celebration, Emerging Technologist Award, Winner 2026
NCWIT Aspirations in Computing Collegiate Award, Grand Prize ($8,500) 2026
CMD IT and ACM Richard Tapia Conference, First Place Graduate Research Poster ($1,500) 2025
Bowers CIS Inspiring Futures Award for mentorship 2025
Cornell Graduate School Exemplary Leadership Award for Early Career Graduate Students 2024
Cadence Technology Award, Women in Technology ($5,000) 2023

Service and Teaching

- **Teaching assistant.** Machine Learning and Reinforcement Learning, Cornell, 2023 to 2024. COS 126 Computer Science, Princeton, 2022 to 2023.
- **Mentoring.** 18 undergraduate, master's, and Ph.D. researchers across robot learning, language models, multi agent learning, and experimental study operations.
- **Reviewer.** ICRA, ACM/IEEE HRI, ACM THRI, AAMAS, NeurIPS, and machine learning workshops, 2023 to 2026.
- **Organizing.** Co-organizer and Outreach Lead, HRI 2027 workshop on long term real world robot deployments. Co-organizer, Black in Robotics session at IROS 2026 Community Building Day.